

EEG Pipeline – Setup Guide

Emotiv EPOC X (14 channels) | MATLAB + EEGLAB | runs entirely offline

What this does

Takes raw EEG recordings, automatically removes noise and artifacts (drift, blinks, muscle, bad electrodes), saves the cleaned data, and produces power spectrum plots from 0–20 Hz for all 14 channels. Nothing is ever uploaded — all processing happens on this computer.

Step 1 · Install MATLAB

MATLAB R2021a or newer, with these two toolboxes:

- Signal Processing Toolbox
- Statistics and Machine Learning Toolbox

Step 2 · Install EEGLAB

Download from [eeglab.org](https://www.sccn.ucsd.edu/eeglab) and unzip it into the **toolboxes** folder inside this project. Then add these four plugins from inside EEGLAB, via *File > Manage EEGLAB extensions*:

- **BIOSIG** — reads EDF/BDF files
- **firfilt** — the filters
- **clean_rawdata** — removes large artifacts (ASR)
- **ICLabel** — identifies blinks and muscle noise

Step 3 · Add your recordings

Put them in the **data\raw** folder. Accepted formats:

- **.edf** or **.bdf** — preferred
- **.csv** — EmotivPRO export

This folder is never modified. Original recordings stay untouched.

Step 4 · Start it

Double-click **1 - START HERE.bat**, or open MATLAB and run:

```
cd('<this folder>')
addpath('src')
cfg = setup_paths();
check_env();
```

check_env() lists anything still missing. If it says “All hard requirements met”, you are ready.

Step 5 · Process everything

```
results = run_pipeline();
```

That is the whole thing. It cleans every recording, saves the cleaned data, plots each spectrum, and builds the three-recording comparison figure.

Where the results appear

figures\	Power spectrum plots, plus the three-recording comparison
results\qc_summary.csv	Open this first. One row per recording, listing every decision made. The usable column: 1 = fine, 0 = too noisy to trust
data\processed\	Cleaned recordings, one .set file each

Adjusting the pipeline. Every filter setting, threshold and on/off switch lives in **src\pipeline_config.m**, each with a comment explaining why it has the value it does. Nothing needs editing anywhere else.

Note on short recordings. ICA and ASR need roughly 30–45 seconds of data to work reliably. On shorter recordings the pipeline skips them and says so, rather than producing a result that cannot be trusted.

Full documentation: **README.md** · folder map: **WHERE_EVERYTHING_IS.md** · detailed install: **docs\SETUP.md**